Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/15/2020
PI Name:	Langer, Robert Sc.D.		
Project Title:	Gastrointestinal Devices for Long-Term In Situ Delivery of Therapeutic Microbes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02142	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Traverso, Carlo M.D., Ph.D. (Brigham and Women's Hospital)		
Grant/Contract No.:	NNX16AO69A-T0107		
Performance Goal No.:			
Performance Goal Text:			
	This proposal addresses research topic 8 (Microbiome based therapies for improving health in spaceflight) through the development of a gastrointestinal (GI) device with long residence time that hosts and administers genetically engineered therapeutic microbes in situ. Genetically engineered therapeutic microbes represent a promising approach to modulating the gut microbiome and enhancing human health. Importantly, this strategy can be designed with molecular specificity and provides substantial advantages over traditional prebiotic and probiotic treatments. In fact, synthetic microbes have already been deployed to deliver molecules against inflammatory bowel diseases (IL-10), over-eating (N-acylethanolamines), bacterial pathogens (bacteriocins), and diabetes (isulinotropic proteins). However, delivery and containment of synthetic microbes remains a substantial challenge to the translation of this strategy to humans.		

Task Description:	Thus far research has focused on strategies to enable synthetic microbes to stably colonize the gut as a means of long-term delivery. However, this goal is problematic both because it places undue requirements on the choice of microbe and because it leads to the permanent introduction of genetic modifications to the natural microbiome. Furthermore, synthetic microbes with an enhanced ability to colonize the human body will be hard to contain and may invade off-target niches. Thus we propose to develop a polymeric device that is retained in the GI tract for an extended time interval (1+ years) to host, protect, and administer genetically modified therapeutic microbes. This strategy decouples the challenge of delivery from the genetic requirements of the therapeutic microbe enabling the immediate translation of available synthetic microbes to enhance health during spaceflight. Importantly, this strategy allows the use of model organisms for which there exists a large body of genetic tools for the robust implementation of therapeutic properties as well as biosafety mechanisms. Additionally, this GI device minimizes personnel, equipment, and space requirements during spaceflight since it can be administered and quality checked during a pre-flight period. This builds from our previous work on gastric-resident devices for the long-term delivery of small-molecule therapeutics. In this proposal we aim to validate the ability of such a device to (1) preserve the viability of microbes, (2) administer the microbes with a targeted regimen, and (3) remain safely in the GI tract in spaceflight-like conditions. We will evaluate these key parameters in vitro and select optimal devices for in depth study in vivo in a large animal model (swine). Success of this project will represent a significant advance in our ability to deploy microbiome therapies based on synthetic microbes both for spaceflight as well as for Earth-based clinical contexts. Additionally, the resulting platform can be extended to incorporate advanced genetic and electronic systems to enable integrated devices with sense and response as well as remote control capabilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Synthetic microbes delivered to the GI tract have the potential to mitigate several risks defined by the Human Research Program, such as altered astronaut immune response, inadequate nutrition, host-microorganism interactions, and adverse cognitive disorders. If successful, our microbe delivery system will enable translation of these synthetic microbe therapies for use during spaceflight. This system will minimize personnel, equipment, and space requirements since it can be administered and quality checked during a pre-flight period. Furthermore, the general class of gastric-resident devices used in this work is a drug delivery modality new to spaceflight. This strategy decouples the challenge of delivery from the genetic requirements of the therapeutic microbe enabling the immediate translation of available synthetic microbes to enhance health during spaceflight. Success of this project will represent a significant advance in our ability to deploy microbiome therapies based on synthetic microbes both for spaceflight as well as for Earth-based clinical contexts. Over the past reporting period we have discovered key materials and developed key methods and devices that enable both our goal of a gastric resident system for microbial delivery as well as the development of future microbe-based systems that require live, stabilized microbial cells. Specifically, we discovered materials that stabilize microbes through the tableting process. Importantly, this enables the microbes to be used directly as stabilized tablets in space with no cold storage. Furthermore they enable us to incorporate them simply into gastric resident systems. Additionally, we developed a new gastric resident system that maximizes and simplifies microbial loading. Finally, we validated this new system for function and safety in our swine large animal model. Together, the ability to stably tablet microbes and incorporate them into GI resident devices sets the stage for developing therapeutic and prophylactic microorganisms that can deliver just-in-time medicines.
Task Progress:	Key findings: During this reporting period we had several key findings towards each of our aims. For Aim 1, we surveyed current off-the-shelf microbial formulations and found them to be inadequate; we found safe materials that stabilize the target microbe over 1000x better than off-the-shelf options; and we found that these new formulations allow tableting and incorporation into devices. For Aim 2, we developed a tablet-compatible gastric resident star system; and we validated the tablet-compatible device in vivo and found it to be safely retained for 25+ days. Impact of key findings: Our findings for Aim 1 represent a significant advance in the development of stable microbial formulation compared to available options at the start of the project. These formulations alone open the door to translation of microbial therapeutics to patients and astronauts. Our findings for Aim 2 represent the development of the first ingestible device compatible with a microbial payload and the in vivo validation significantly de-risks this approach to microbial delivery. Results: Overall, our work over the previous reporting period delivered the following results: (1) a stable microbial formulations compatible with tableting, (2) an optimized method and software for discovering new protective formulations under additional stress conditions, (3) a new tablet-compatible gastric resident device, (4) validation of device safety and retention in vivo.
Bibliography Type:	Description: (Last Updated: 05/19/2020)
Articles in Peer-reviewed Journals	Jimenez M, Langer R, Traverso G. "Microbial therapeutics: New opportunities for drug delivery." J Exp Med. 2019 May 6;216(5):1005-9. https://doi.org/10.1084/jem.20190609 ; PMID: 31028093; PMCID: PMC6504217 , May-2019
Awards	Jimenez M. "Translational Research Institute for Space Health Postdoctoral Fellowship, June 2019." Jun-2019
Awards	Langer R. "Alpha Omega Dental Fraternity Achievement Medal Award, October 2018." Oct-2018
Awards	Langer R. "Dreyfus Prize in the Chemical Sciences, September 2019." Sep-2019
Awards	Langer R. "Elected International Fellow of the Royal Society of Canada, December 2018." Dec-2018
Awards	Langer R. "Honorary Degree, Columbia University, May 2019." May-2019
Awards	Langer R. "Hope Funds for Cancer Award of Excellence in Basic Sciences, September 2019." Sep-2019

Awards	Langer R. "Inducted into Hall of Fame, Advanced Materials, November 2018." Nov-2018
Awards	Langer R. "Named Number 1 Translational Researcher in the World (Nature Biotechnology), December 2018." Dec-2018
Awards	Langer R. "National Library of Medicine (Friends) Distinguished Medical Science Award, September 2019." Sep-2019